

Unit Speed Reparametrization

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1. Example.

Consider the curves

$$\gamma(t) = (t, t^2), \quad 0 \leq t \leq 2.$$

$$\bar{\gamma}(s) = (s-1, (s-1)^2), \quad 1 \leq s \leq 3.$$

Both curves have the same image and are related by the equation

$$\bar{\gamma}(s) = \gamma(g(s))$$

where $g : (1, 3) \rightarrow (0, 2)$ is given by

$$g(s) = s - 1.$$

One could think of γ and $\bar{\gamma}$ as representing the trajectory of an object as seen by two observers, one whose watch is a second slower than that of the other. The map g compensates for the difference.

2. Definition.

Let

$$\bar{\gamma} : (s_0, s_1) \rightarrow \mathbb{R}^3$$

and

$$\gamma : (t_0, t_1) \rightarrow \mathbb{R}^3$$

be curves γ is called a *smooth reparametrization* of γ if there exists a function $g : (s_0, s_1) \rightarrow (t_0, t_1)$ such that

- (a) g is a smooth (i.e., infinitely differentiable) bijection.
- (b) g^{-1} is also smooth
- (c) $\bar{\gamma} = \gamma \circ g$

3. Remarks.

- (a) The second condition in definition (2) is not redundant. For example, $g(s) = s^3$ is a smooth bijection but not so its inverse which is not differentiable at $t = 0$.

- (b) Since $g^{-1} \circ g(s) = s$ it follows by the chain rule and that g' is non vanishing. Hence g must be either strictly increasing or strictly decreasing and hence map intervals to intervals. Moreover

$$\{\bar{\gamma}(s_0), \bar{\gamma}(s_1)\} = \{\gamma(t_0), \gamma(t_1)\}.$$

- (c) If $\bar{\gamma}$ is a reparametrization of γ then γ is a reparametrization of $\bar{\gamma}$.

4. Definition.

A curve γ is regular if γ' is non vanishing.

5. Exercise.

Suppose that $\bar{\gamma}$ is a reparametrization of γ . Show that:

- (a) γ and $\bar{\gamma}$ have the same image.
- (b) If γ is regular, then so is $\bar{\gamma}$.
- (c) the tangent line to $\bar{\gamma}$ at s and the tangent line to γ at $g(s)$ are the same line.

6. Definition.

A reparametrization is *unit speed* if

$$\|\bar{\gamma}'(s)\| = 1, \text{ for all } s \in (s_0, s_1).$$

7. Exercise.

If γ is a unit speed curve and $\gamma'' \neq 0$, show that $\gamma' \perp \gamma''$.

8. Remark.

Unit speed is good for standardization. If a physicist, living in a Newtonian universe, were to define the curvature of a path as the acceleration scalar of a unit mass travelling at unit speed along the path, he would arrive at the mathematicians definition of curvature.

¹<https://pennance.us> Reference: Manfredo P. do Carmo. Differential Geometry of Curves and Surfaces.

9. Claim.

The following are equivalent:

- (a) γ is regular.
- (b) γ has a unit speed reparametrization.

Proof.

Suppose that γ is regular. The basic idea is that if distance travelled is used as the measure of time then speed becomes 1. Specifically, pick a point t_0 in the domain of γ and consider the *signed arclength function* defined for all t in the domain of γ by

$$s(t) = \int_{t_0}^t \|\gamma'(\tau)\| d\tau.$$

Since γ is regular, $s'(t) = \|\gamma'(t)\| \neq 0$. The function s is smooth with non zero derivative. The inverse function theorem then guarantees that s^{-1} is also smooth. This means that $\bar{\gamma} = \gamma \circ s^{-1}$ is a reparametrization of γ . Finally

$$\begin{aligned} \bar{\gamma}' &= \gamma' \circ (s^{-1})s^{-1'} \\ &= \frac{\gamma' \circ (s^{-1})}{s' \circ s^{-1}} \\ &= \frac{\gamma' \circ (s^{-1})}{\|\gamma' \circ s^{-1}\|}, \end{aligned}$$

and it follows that $\|\bar{\gamma}'(s)\| = 1$ proving that γ has a unit speed reparametrization, indeed, one for each choice of t_0 .

Conversely, suppose that γ has a unit speed reparametrization $\bar{\gamma}$.

$$\bar{\gamma} = \gamma \circ g.$$

Then, for all s

$$\begin{aligned} 1 &= \|\bar{\gamma}'\| \\ &= \|\gamma'(g(s))\| \cdot |g'(s)|. \end{aligned}$$

It follows that $\gamma' \neq 0$.

10. Example.

Let $f(x) = \sqrt{1-x^2}$, $-1 < x < 1$. The graph G_f of f is the image of an infinite number of parametrized curves. For example,

$$\begin{aligned} \gamma(t) &= (t, \sqrt{1-t^2}), \quad -1 < t < 1 \\ \beta(s) &= (-s, \sqrt{1-s^2}), \quad -1 < s < 1 \\ \alpha(s) &= (s^3, \sqrt{1-s^6}), \quad -1 < s < 1, \end{aligned}$$

all have image G_f . Notice that β is a smooth reparametrization of γ . In fact,

$$\beta(s) = \gamma \circ g(s)$$

where $g : (-1, 1) \rightarrow (-1, 1)$ is given by $g(s) = -s$. Since

$$\gamma'(t) = \left(1, \frac{-t}{\sqrt{1-t^2}}\right)$$

is non vanishing, γ is regular. Because smooth reparametrizations preserves regularity, β is also regular. On the other hand α is not regular and therefore has no unit speed reparametrization.

11. Example.

Find a unit speed reparametrization of the curve

$$\gamma(t) = (t, \sqrt{1-t^2}), \quad -1 < t < 1.$$

Recall that a unit speed reparametrization of γ can be obtained by computing the signed arc-length starting from some point $\gamma(t_0)$. If $t_0 = 0$ the arc-length from $\gamma(0)$ is:

$$\begin{aligned} s &= \int_0^t |\gamma'(\tau)| d\tau \\ &= \int_0^t \frac{1}{\sqrt{1-\tau^2}} d\tau \\ &= \sin^{-1}(t), \quad -1 < t < 1. \end{aligned}$$

The inverse function $g(s) = \sin(s)$, $-\pi/2 < s < \pi/2$ when composed

with γ yields a unit speed parametrized curve:

$$\begin{aligned}\tilde{\gamma}(s) &= \gamma \circ g(s) \\ &= \left(\sin s, \sqrt{1 - \sin^2 s} \right) \\ &= (\sin s, \cos s), \quad -\frac{\pi}{2} < s < \frac{\pi}{2}.\end{aligned}$$

12. Remark.

Each choice of t_0 yields a different unit speed curve. For example, choosing $t_0 = -\frac{1}{2}$ in example (11) gives the arc-length function:

$$\begin{aligned}s(t) &= \int_{-\frac{1}{2}}^t |\gamma'(\tau)| d\tau \\ &= \int_{-\frac{1}{2}}^t \frac{1}{\sqrt{1 - \tau^2}} d\tau \\ &= \sin^{-1}(t) - \sin^{-1}\left(-\frac{1}{2}\right).\end{aligned}$$

In this case the inverse function $t = g(s)$ is given by

$$t = \sin\left(s - \frac{\pi}{6}\right), \quad -\frac{\pi}{3} < s < \frac{2\pi}{3}$$

and the corresponding unit speed reparametrization is

$$\begin{aligned}\tilde{\gamma}(s) &= \gamma \circ g(s) \\ &= \left(\sin\left(s - \frac{\pi}{6}\right), \cos\left(s - \frac{\pi}{6}\right) \right).\end{aligned}$$

13. Claim (Invariance of arclength under reparametrization.)

Let $\gamma : [t_0, t_1] \rightarrow \mathbb{R}^3$ and $\bar{\gamma} : [s_0, s_1]$ be curves in $\mathbb{R}^3$. If $\bar{\gamma}$ is a reparametrization of γ then

$$\int_{s_0}^{s_1} \|\bar{\gamma}'(s)\| ds = \int_{t_0}^{t_1} \|\gamma'(t)\| dt.$$

Proof.

Since $\bar{\gamma}$ is a reparametrization of γ there exists a smooth bijection g such that

$$\gamma(s) = \gamma \circ g(s).$$

Recall that g is either strictly increasing or strictly decreasing. Suppose that g is decreasing. Then,

$$\begin{aligned}\int_{s_0}^{s_1} \|\bar{\gamma}'(s)\| ds &= \int_{s_0}^{s_1} \|(\gamma \circ g)'(s)\| ds \\ &= \int_{s_0}^{s_1} \|\gamma' \circ g(s)\| \cdot |g'(s)| ds \\ &= - \int_{s_0}^{s_1} \|\gamma' \circ g(s)\| \cdot g'(s) ds \\ &= - \int_{g(s_0)}^{g(s_1)} \|\gamma'(t)\| dt \\ &= - \int_{t_1}^{t_0} \|\gamma'(t)\| dt \\ &= \int_{t_0}^{t_1} \|\gamma'(t)\| dt\end{aligned}$$

A similar argument works for the case g increasing.

Exercises

1. Find a parametrization $\gamma(t)$ of the upper half of the unit circle with the property that $\gamma(0) = (1, 0)$ and $\gamma(1) = (-1, 0)$.

2. Find a parametrization

$$\alpha(t), \quad 0 \leq t \leq 1$$

of the upper half of the unit circle with the property that $\alpha(0) = (-1, 0)$ and $\alpha(1) = (1, 0)$

3. Let $\gamma : [a, b] \rightarrow \mathbb{R}^3$. Find a reparametrization $\hat{\gamma}$ such that $\hat{\gamma}(a) = b$ and $\hat{\gamma}(b) = a$.
4. Reparametrize by arclength the cylindrical helix given by

$$\gamma(t) = (a \cos t, a \sin t, bt).$$

5. Find a unit speed reparametrization of

the circle of radius $r > 0$ and center (h, k) .

6. Let

$$f(x) = \frac{x^3}{6} + \frac{1}{2x}, \quad 1 \leq x \leq 3.$$

Parametrize the graph of f and hence find the arclength ℓ between the points $(1, f(1))$ and $(3, f(3))$.

7. Let γ, γ' be parametrized curves. Define $\gamma \sim \gamma'$ if γ' is a smooth reparametrization of γ . Show that $\sim$ is an equivalence relation. What can be said concerning the equivalence classes?
8. Let $\alpha = \gamma \circ g$ and $\beta = \gamma \circ h$ be two unit speed reparametrizations of a curve γ . How must g and h be related? Prove your answer.